

Chirag Hegde

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SKILLS

- Languages: Python, R, C, Java, CSS, HTML, JavaScript, MATLAB, NodeJS, ReactJS, Kotlin, Tableau, Ruby, Vue.js, Chart.js
- Database and Operating Systems: PostgreSQL, MySQL, MongoDB, Ubuntu
- Tools/Frameworks: GIT, Postman API, AWS, Hadoop, Docker, Swagger
- Certifications: Certified as an AWS Academy Graduate - AWS Academy Cloud Foundations from ICT Academy, Palo Alto Networks Academy Cybersecurity Foundation (Coursera), HTML, CSS, and JavaScript for Web Developers (Coursera), Cloud Practitioner (Course)

PROFESSIONAL EXPERIENCE

Teaching Assistant for CSC 565: Graph Theory, NCSU

Aug – Dec 2024

- Supported graduate-level instruction by assisting with coursework design, grading assignments, and conducting problem-solving sessions.
- Held weekly office hours to help students master topics such as graph traversal, connectivity, planarity, and network flows.
- Collaborated with the course instructor to create assessment rubrics and guide project-based learning.

Software Engineer Intern, TetherFi, Bengaluru, India

May – Nov 2022

- Developed an MVC Asp.net Core web application for an HRIS system focused on recruitment, leveraging Entity Framework for ORM capabilities and integrating Identity for user authentication as part of the core product development team.
- Engineered a chatbot that retrieves institutional documents from any server within the organization, utilizing RESTful APIs for data fetching and JSON for data interchange to enhance precision and efficiency.
- Employing Swagger for API documentation to optimize performance and scalability.
- Designed an advanced NLP model using TensorFlow for machine learning algorithms and BERT for natural language understanding, significantly improving the chatbot's ability to provide accurate and contextually relevant document retrieval.

Software Engineer Intern, SoftLink, Mumbai, India

Feb – Dec 2021

- Supported the successful implementation of a digital transformation initiative, utilizing Azure cloud services and SAP ERP solutions to overhaul logistics operations.
- Played a key role in a cross-functional team that adopted machine learning algorithms and integrated IoT devices using MQTT protocols, resulting in a marked increase in operational efficiency and real-time data analytics.

PROJECTS

Student Attentiveness Evaluation in Classroom Using Face Recognition and Machine Learning

- Developed a real-time attentiveness checker using facial cues (head pose, gaze direction) and student-specific CNN models to classify engagement.
- Integrated Google OAuth and Classroom API to automatically load class assignments and personalize monitoring per student.
- Built a browser-based frontend and Flask backend where OpenCV captures frames and TensorFlow classifies attention at 1 FPS, with results stored in MongoDB.
- Technologies used: Python, Flask, OpenCV, TensorFlow, Google OAuth API, MongoDB, JavaScript

E-Commerce Storefront with Stripe Integration

- Built a full-featured e-commerce platform for digital product sales, supporting creators and businesses in monetizing content.
- Developed a responsive, user-friendly UI and implemented seamless payment processing with Stripe for secure transactions.
- Integrated secure payment flows using the Stripe API, supporting one-time and recurring payments.
- Technologies used: Next.js, TypeScript, Stripe API

Sign Language Detection Using Deep Learning

- Utilized deep learning to recognize and interpret sign language with 95% accuracy using a Convolutional Neural Network (CNN) optimized through hyperparameter tuning.
- Technologies used: Python, TensorFlow, CNN, Kaggle datasets

PersonalDoctor Ai-LLM-based-Healthcare

- Architected an LLM-powered healthcare chatbot combining a fine-tuned RoBERTa model for medical question answering with a scikit-learn Random Forest classifier for symptom-based disease prediction, resulting in a highly efficient solution for symptom checking and accurate medical query responses.
- Built a custom medical Q&A pipeline by scraping authoritative health websites to create a domain-specific dataset and fine-tuning the RoBERTa LLM using HuggingFace Transformers and Fast.ai, enabling precise, context-rich answers to patient queries.
- Engineered an end-to-end data pipeline that scraped and cleaned medical content from Healthline to create a 6.8K-sample Q&A corpus, and processed a Kaggle disease-symptom database for training the Random Forest model.
- Technologies used: Python, HuggingFace Transformers (RoBERTa), Fast.ai, PyTorch, scikit-learn , Flask, Gradio, BeautifulSoup, pandas, NumPy

EDUCATION

North Carolina State University - Raleigh, United States

Aug 2023 - May 2025

- Master of Computer Science

GPA: 3.63/4.0

- Coursework: Automated Learning and Data Analysis, Artificial Intelligence, Social Computing and Decentralized AI, Graph Theory, Cloud Computing, Software Engineering, Neural Network and Deep Learning, Object-oriented Design and Development, Design and Analysis of Algorithms, Parallel Systems

A.P. Shah Institute of Technology - Thane, India

Aug 2019 - May 2023

- Bachelor of Engineering in Computer Engineering

CGPA :8.9/10

- Coursework: Data Analytics, Machine Intelligence, AI, Cloud Computing, Computer Networks, OS, Big Data, Network Analysis and Mining, Compiler Design, Microprocessors, Software and Systems Performance